

Understanding PCA: Covariance, Eigenvectors, and Eigenvalues

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1. Introduction to PCA

Principal Component Analysis (PCA) is a dimensionality reduction technique that transforms a high-dimensional dataset into a new coordinate system such that the greatest variance by any projection of the data lies on the first coordinate (called the first principal component), the second greatest variance on the second coordinate, and so on.

2. The Role of Covariance Matrix

The covariance matrix captures how features vary with each other:

$$\Sigma = \frac{1}{n-1} X^T X$$

Where:

- X is the data matrix with features centered around zero
- Σ is a symmetric matrix containing variances along the diagonal and covariances off-diagonal

Example Dataset (2D)

Consider the following simple dataset:

```
X = [ [2.5, 2.4],  
      [0.5, 0.7],  
      [2.2, 2.9],  
      [1.9, 2.2],  
      [3.1, 3.0],  
      [2.3, 2.7],  
      [2.0, 1.6],  
      [1.0, 1.1],  
      [1.5, 1.6],  
      [1.1, 0.9] ]
```

The scatter plot of this data shows it is mostly aligned along a diagonal axis:

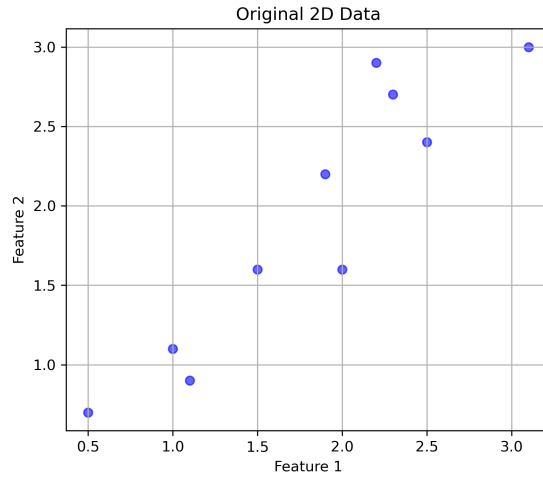


Figure 1: Original 2D data plot

3. Eigenvectors and Eigenvalues

To find the principal components, we solve the equation:

$$\Sigma \vec{v} = \lambda \vec{v}$$

Where:

- $\vec{v}$ is the eigenvector (direction)
- λ is the eigenvalue (magnitude of variance in that direction)

Interpretation:

- The largest eigenvector points in the direction of maximum data variance.
- Each eigenvector is orthogonal (perpendicular) to the others.
- Eigenvalues represent the amount of variance explained by each eigenvector.

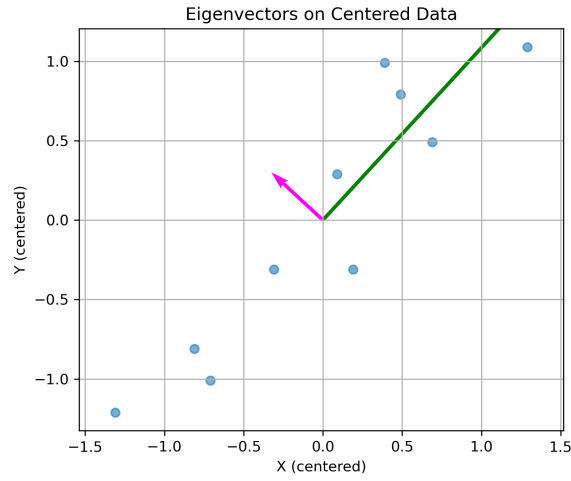


Figure 2: Eigenvectors (green and magenta) overlaid on data

4. Transforming the Data

We project the data onto the new basis formed by the principal components (eigenvectors):

$$X_{\text{new}} = X \cdot W$$

Where W is the matrix of eigenvectors. The result is the same data, now aligned along the principal axes.

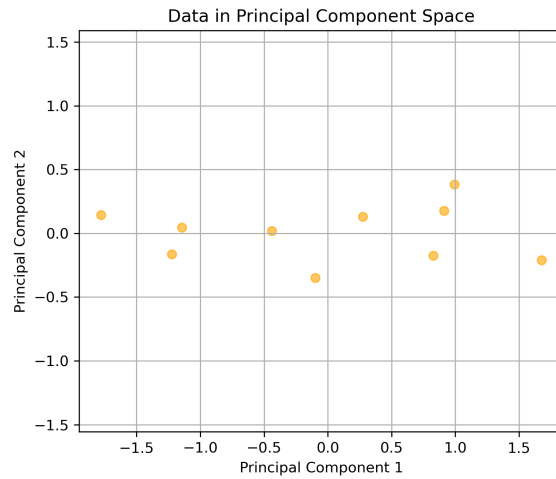


Figure 3: Data transformed into principal component space

5. Special Case: Diagonal Covariance Matrix

If the covariance matrix is diagonal:

$$\Sigma = \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix}$$

Then:

- Features are uncorrelated.
- The diagonal values (variances) are equal to the eigenvalues.
- PCA does not need to rotate the data.

6. Summary

- PCA finds new axes (eigenvectors) that best explain the spread (variance) in the data.
- These axes are ranked by their corresponding eigenvalues.
- The covariance matrix is the starting point, capturing relationships between features.
- PCA helps reduce dimensions while preserving as much variance as possible.